

Emerging AI news / technologies

- **DeepSeek's Engrams:** DeepSeek introduced the "Engram" via a research paper and github repository this January.
- The general philosophy behind it is basically; what if the AI model has a fast memory module it can look up to for common patterns, so that the main brain can stop wasting time and resources on simple stuff and focus on real thinking.
- The Engram is an architectural innovation designed to make LLMs more efficient by replacing repetitive, compute-heavy, early-layer calculations with a fast, hash-based memory lookup for common patterns. By storing unchanging knowledge in a "second brain" of billions of parameters, it offloads factual recall to improve reasoning in complex tasks, reducing costs. It is a conditional memory module (named "Conditional Memory via Scalable Lookup") designed to allow LLMs to separate static knowledge storage from dynamic reasoning, reducing reliance on GPU memory. It is expected to be a foundational model for the upcoming **DeepSeek V4** model. It allows for constant-time knowledge retrieval from large contexts by utilizing system RAM (DRAM) rather than the more expensive VRAM. constant-time being an operation that takes consistent and predictable time regardless of the magnitude of input.

AI Applications in Healthcare

Problems AI / successful innovations	Problems AI is beginning to solve / Partly solved	Problems AI has not solved / Not presently Feasible
General Medical Image Analysis: AI systems now match or exceed specialist accuracy in interpreting chest X-rays, mammograms, and retinal scans for disease detection.	Personalized Medication: AI can suggest treatments based on patient data, but recommendations still require extensive physician oversight and validation due to concerns about accuracy and contextual understanding.	Data Diversity and Equitable Healthcare Delivery: AI cannot yet overcome systemic biases in training data, often perpetuating existing healthcare disparities across different populations, socioeconomic groups, and geographic regions. This is mostly due to lack of data diversity.
Diabetic Retinopathy Screening: AI systems autonomously	Predictive Medication: AI analyzes patient genetics, medical history,	Transparent "Explainable AI": Current AI systems

analyze retinal images to detect diabetic retinopathy, age-related macular degeneration, and glaucoma with accuracy matching or exceeding specialist ophthalmologists, enabling early intervention especially in underserved areas with limited access to eye care specialists.	and biomarkers to predict which medications will be most effective and which may cause adverse reactions, enabling personalized treatment plans that improve therapeutic outcomes while reducing trial-and-error prescribing, but recommendations still require extensive physician oversight and validation due to concerns about accuracy and contextual understanding.	struggle to explain their reasoning in ways clinicians can understand and validate, making it difficult to trust recommendations in critical situations.
Stroke Detection: AI algorithms analyze CT and MRI brain scans to rapidly identify stroke type (ischemic vs. hemorrhagic), measure clot size, and alert specialists within minutes of imaging, reducing diagnosis time from hours to seconds and enabling faster treatment that significantly improves patient outcomes during the critical hours.	Drug Research: AI is reducing drug discovery timelines from years to months in research settings, but translation to approved therapies and market availability remains in early stages.	Privacy and Data security / Data Privacy Protection: AI models trained on clinical data can unintentionally reproduce identifiable patient information, violating HIPAA and GDPR regulations despite security measures.
Personal and Remote Patient Monitoring: Wearable devices with AI successfully track vital signs and detect anomalies like arrhythmias, enabling early intervention for chronic conditions.		Clinical Accountability and Liability: No clear legal framework exists for determining responsibility when AI-assisted diagnosis leads to harm, whether it falls on the clinician, hospital, or AI developer.
		Overreliance On AI/AI replacing Clinical Human

		Judgement: AI is still yet to replicate the nuanced, contextual decision-making that experienced physicians bring to complex cases, especially those involving rare conditions or unusual presentations.
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AI Applications in Agriculture

Problems AI has solved / Successful Innovations	Problems AI Is beginning to Solve / Partly Solved	Problems AI Has Not Solved / Not Presently Feasible
Precision Farming: AI optimizes seed selection and planting strategies based on soil composition and historical climate data, enabling field-level precision down to the nearest meter or individual plant.	Autonomous Farm Equipment: Self-driving tractors and harvesters from companies like John Deere and CNH Industrial operate with minimal human intervention, but widespread adoption is limited by high capital costs and the need for farm-specific customization.	Transparent Explainable AI: Current AI systems struggle to explain their reasoning in ways clinicians can understand and validate, making it difficult to trust recommendations in critical situations.
Satellite-Based Crop Monitoring: AI analyzes satellite imagery and vegetation indices like NDVI to assess crop health, detect nutrient deficiencies, and identify disease outbreaks early with high precision across millions of hectares.		Data Quality and Availability: Agriculture lacks high-quality, diverse datasets representing different farm types, marginalized regions, and crop varieties. Most training data comes from large industrial farms, creating biased and fragmented AI models.
Supply chain		

Optimization: AI streamlines food distribution by forecasting demand, minimizing losses, and improving logistics efficiency, reducing waste in the supply chain by analyzing real-time market and transportation data.		
Yield optimization: AI analyzes historical yield data, soil conditions, weather patterns, and crop health metrics to recommend optimal planting densities, fertilizer application rates, irrigation schedules, and harvest timing. This enables farmers to maximize crop production while minimizing resource waste and input costs.		
Demand Forecasting: AI analyzes consumer buying patterns, seasonal trends, weather forecasts, and market dynamics to predict crop demand weeks or months in advance, enabling farmers to plant the right quantities of specific crops, reduce overproduction waste, optimize pricing strategies, and coordinate with distributors to match supply with market needs.		

Overall, AI has been successfully applied to many aspects of healthcare and agriculture, with a number of other solutions still being developed for different problems in these fields. On the other side, some AI solutions appear to not be presently viable,

mostly due to consumer protection policy regulations like GDPR and HIPAA, alongside integration issues.